

ABDULLAH KHAN

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EDUCATION

University of Waterloo

Bachelor of Computer Science (BCS)

Waterloo, ON

Expected 2029

Wilfrid Laurier University (Double Degree)

Bachelor of Business Administration (BBA)

Waterloo, ON

Expected 2029

TECHNICAL SKILLS

Languages: Python, JavaScript, TypeScript, Java, SQL, C++

Frameworks & Libraries: React, Node.js, Express.js, FastAPI, Puppeteer, Tailwind CSS, Chart.js

Tools & Technologies: Git/GitHub, GitLab, PostgreSQL, SQLite, Docker, AWS (Lambda, S3, RDS, ECS), Grafana, Graylog, Vercel

EXPERIENCE

Software Engineering Intern

LeapAP Inc.

Aurora, ON

May 2026 – Aug. 2026

- Cut new-integration build time from **3+ hours to under 10 minutes** by designing a generic Node.js/Puppeteer scraper framework that reduces each integration to 3 config changes; it is now the base for 11 production scrapers.
- Expanded billing data coverage by shipping **32+ scrapers and 28+ parsers** for utility, telecom, and waste portals.
- Restored data collection across 100+ billing portals, some with up to 2,700 accounts, by resolving **110+ production failures** (selector changes, MFA parsing, CAPTCHAs) using Grafana and Graylog.
- Improved error-report accuracy as sole owner of a **biweekly audit** of false “account not found” and “bad credentials” errors, surfacing 10+ misclassified failures so error reporting stays trustworthy across all integrations.
- Closed **185+ tickets** in 4 months while deploying to production multiple times daily via GitLab.

Software Developer

PixelsBoost

Milton, ON

Sep. 2025 – Dec. 2025

- Delivered **5 full-stack client websites** to production using React, HTML/CSS, and JavaScript, implementing dynamic contact forms, interactive image galleries, and mobile-responsive navigation for small-business clients.
- Integrated **3 third-party APIs**, including Stripe for payment processing, Google Maps for geolocation, and SendGrid for email automation, with error handling and retry logic to keep checkout and contact flows reliable.
- Raised Google Lighthouse scores from **68 to 87** through WebP image compression, lazy loading, and Cloudflare CDN caching, speeding up page loads on image-heavy client sites.
- Managed **5 concurrent client projects** using Git/GitHub feature-branch workflows, collaborating with 2 designers to turn mockups into production-ready pages.

PROJECTS

BookPulse | [Live](#) | [GitHub](#) | React, TypeScript, FastAPI, PostgreSQL, Docker, AWS, Chart.js

- Built and deployed a full-stack **book price intelligence platform** with a React/TypeScript frontend on Vercel and a FastAPI backend on Render, connected to AWS RDS PostgreSQL to track 100+ books with rarity and demand scoring.
- Architected a serverless **AWS Lambda** ingestion pipeline triggered hourly by EventBridge, storing raw book data in S3 and computing price change metrics, demand scores, and rarity rankings across 5 categories.
- Containerized the backend with **Docker** and deployed it to AWS ECS Fargate, configuring IAM roles, security groups, and CORS middleware for secure communication between the Vercel frontend and cloud backend.
- Built a live dashboard with Chart.js visualizations, a top-movers table, and a rarity leaderboard; fixed a **React StrictMode** bug where double-invoked effects corrupted useRef state on mount.

Attack Surface Growth Simulator | [Live](#) | [GitHub](#) | Python, React, FastAPI, SQLAlchemy, NumPy, SQLite, Recharts

- Built a full-stack security risk modeling tool that scores attack surface (**0–100**) across **5 threat categories** by normalizing system metrics (endpoints, users, MFA adoption, vulnerabilities) and generating ranked driver breakdowns.
- Developed a **FastAPI** backend with Pydantic validation that computes quadratic risk functions and uses NumPy derivatives to identify growth zones where system complexity creates disproportionate security exposure.
- Designed a **SQLite** database with the SQLAlchemy ORM to persist assessment configurations and results, letting users compare security scenarios and track how architecture changes affect overall risk.
- Built an interactive **React** dashboard with Recharts that visualizes risk curves, growth rates, and danger zones as users adjust system parameters.